Solutions

Algebraic problems

Question 8.1:

(a) Let's use $u = 5x+6$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 5x+6 \implies x = \frac{1}{5}(u-6) \implies \frac{dx}{du} = \frac{1}{5} \implies dx = \frac{1}{5}du$$

Use these to make the substitution:

$$\int 4\,(5x+6)^3\,dx \implies \frac{4}{5}\int u^3 du$$

Now we can evaluate using the basic laws of integration:

$$\frac{4}{5}\int u^3 du = \frac{4}{5}\times\frac{1}{4}\times u^4 + c$$

and substituting $u = 5x+6$ back in gives us:

$$y = \frac{1}{5}u^4 + c$$

You can differentiate using the chain rule to verify that this works.

$$u = x-2 \implies x = u+2 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x-2)^3\,dx \implies \int u^3 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^3 du = \frac{1}{4}u^4 + c$$

and substituting $u = x-2$ back in gives us:

$$y = \frac{1}{4}(x-2)^4 + c$$

You can differentiate using the chain rule to verify that this works.

(c) Let's use $u = x+5$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+5 \implies x = u-5 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x+5)^4\,dx \implies \int u^4 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^4 du = \frac{1}{5}u^5 + c$$

and substituting $u = x+5$ back in gives us:

$$y = \frac{1}{5}(x+5)^5 + c$$

You can differentiate using the chain rule to verify that this works.

(d) Let's use $u = 1-2x$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 1-2x \implies x = \frac{1}{2}(1-u) \implies \frac{dx}{du} = -\frac{1}{2} \implies dx = -\frac{1}{2}du$$

Use these to make the substitution:

$$\int (1-2x)^7 dx \implies -\frac{1}{2}\int u^7 du$$

Now we can evaluate using the basic laws of integration:

$$-\frac{1}{2}\int u^7 du = -\frac{1}{16}u^8 + c$$

and substituting $u = 1-2x$ back in gives us:

$$y = -\frac{1}{16}(1-2x)^8 + c$$

You can differentiate using the chain rule to verify that this works.

Question 8.2:

(a) Let's continue from Question 8.1(a) where we used a substitution of $u = 5x+6$. Find the new limits:

$$\text{When } x = 0,\ u = 5\times 0 + 6 = 6$$
$$\text{When } x = 1,\ u = 5\times 1 + 6 = 11$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 4(5x+6)^3 dx &= \frac{4}{5}\int_6^{11} u^3 du \\ &= \frac{4}{5}\left[\frac{1}{4}u^4\right]_6^{11} \\ &= \frac{1}{5}(11^4 - 6^4) = 2669\end{aligned}$$

(b) Let's continue from Question 8.1(b) where we used a substitution of $u = x - 2$. Find the new limits:

$$\text{When } x = 0,\ u = 0 - 2 = -2$$
$$\text{When } x = 1,\ u = 1 - 2 = -1$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 (x-2)^3\,dx &= \int_{-2}^{-1} u^3 du \\ &= \left[\frac{1}{4}u^4\right]_{-2}^{-1} \\ &= \frac{1}{4}((-1)^4 - (-2)^4) = -3.75\end{aligned}$$

(c) Let's continue from Question 8.1(c) where we used a substitution of $u = x + 5$. Find the new limits:

$$\text{When } x = 0,\ u = 0 + 5 = 5$$
$$\text{When } x = 1,\ u = 1 + 5 = 6$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 (x+5)^4\,dxdx &= \int_5^6 u^4 du \\ &= \left[\frac{1}{5}u^5\right]_5^6 \\ &= \frac{1}{5}(6^5 - 5^5) = 930.2\end{aligned}$$

(d) Let's continue from Question 8.1(d) where we used a substitution of $u = 1 - 2x$. Find the new limits:

$$\text{When } x = -1,\ u = 1 - 2 \times (-1) = 3$$
$$\text{When } x = 1,\ u = 1 - 2 \times 1 = -1$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-1}^1 (1-2x)^7\,dx &= -\frac{1}{2}\int_3^{-1} u^7 du \\ &= -\frac{1}{16}\left[u^8\right]_3^{-1} \\ &= \frac{1}{16}(3^8 - 1^8) = 410\end{aligned}$$

Question 8.3: Using *integration by substitution*, find the indefinite integral of the following:

(a) Let's use $u = 7x - 3$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 7x - 3 \implies x = \frac{1}{7}(u+3) \implies \frac{dx}{du} = \frac{1}{7} \implies dx = \frac{1}{7}du$$

Use these to make the substitution:

$$\int \sin(7x-3)\,dx \implies \frac{1}{7}\int \sin(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{7}\int \sin(u)\,du = \frac{1}{7}\sin(u) + c$$

and substituting $u = 7x - 3$ back in gives us:

$$y = \frac{1}{7}\sin(7x-3) + c$$

You can differentiate using the chain rule to verify that this works.

(b) Let's use $u = 5x + 2$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 5x + 2 \implies x = \frac{1}{5}(u-2) \implies \frac{dx}{du} = \frac{1}{5} \implies dx = \frac{1}{5}du$$

Use these to make the substitution:

$$\int \exp(5x+2)\,dx \implies \frac{1}{5}\int \exp(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{5}\int \exp(u)\,du = \frac{1}{5}\exp(u) + c$$

and substituting $u = 5x + 2$ back in gives us:

$$y = \frac{1}{5}\exp(5x+2) + c$$

You can differentiate using the chain rule to verify that this works.

(c) Let's use $u = 1 - x$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 1 - x \implies x = 1 - u \implies \frac{dx}{du} = -1 \implies dx = -du$$

Use these to make the substitution:

$$\int \cos(1-x)\,dx \implies -\int \cos(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$-\int \cos(u)\,du = -\sin(u) + c$$

and substituting $u = 1 - x$ back in gives us:

$$y = -\sin(1-x) + c$$

(d) Let's use $u = 7x + 5$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 7x+5 \implies x = \frac{1}{7}(u-5) \implies \frac{dx}{du} = \frac{1}{7} \implies dx = \frac{1}{7}du$$

Use these to make the substitution:

$$\int \frac{1}{7x+5}dx \implies \frac{1}{7}\int \frac{1}{u}du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{7}\int \frac{1}{u}du = \frac{1}{7}\ln(u) + c$$

and substituting $u = 7x + 5$ back in gives us:

$$y = \frac{1}{7}\ln(7x+5) + c$$

Question 8.4: Using *integration by substitution*, find the definite integral of the following:

(a) Let's continue from Question 8.3(a) where we used a substitution of $u = 7x - 3$. Find the new limits:

$$\text{When } x = 0,\ u = 7\times 0 - 3 = 4$$
$$\text{When } x = \frac{\pi}{2},\ u = 7\times\frac{\pi}{2} - 3$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^{\pi/2} \sin(7x-3)\,dx &= \frac{1}{7}\int_4^{\frac{7\pi}{2}-3} \sin(u)\,du \\ &= -\frac{1}{7}\left[\sin(u)\right]_4^{\frac{7\pi}{2}-3} \\ &= \frac{1}{7}\left(\sin\left(\frac{7\pi}{2}-3\right) - \sin(4)\right) = 0.25\end{aligned}$$

(b) Let's continue from Question 8.3(b) where we used a substitution of $u = 5x + 2$. Find the new limits:

$$\text{When } x = 0,\ u = 5\times 0 + 2 = 2$$
$$\text{When } x = 1,\ u = 5\times 1 + 2 = 7$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 \exp(5x+2)\,dx &= \frac{1}{5}\int_2^7 \exp(u)\,du \\ &= \frac{1}{5}\left[\exp(u)\right]_2^7 \\ &= \frac{1}{5}(\exp(7)-\exp(2)) = 217.849\end{aligned}$$

(c) Let's continue from Question 8.3(c) where we used a substitution of $u = 1-x$. Find the new limits:

$$\begin{aligned}&\text{When } x = -\frac{\pi}{4},\ u = 1-(-\frac{\pi}{4}) == 1+\frac{\pi}{4} \\ &\text{When } x = \frac{\pi}{4},\ u = 1-\frac{\pi}{4} = 1-\frac{\pi}{4}\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-\pi/4}^{\pi/4} \cos(1-x)\,dx &= -\int_{1+\frac{\pi}{4}}^{1-\frac{\pi}{4}} \sin(u)du \\ &= -\left[\sin(u)\right]_{1+\frac{\pi}{4}}^{1-\frac{\pi}{4}} \\ &= -\left(\sin\left(1-\frac{\pi}{4}\right)-\sin\left(1+\frac{\pi}{4}\right)\right) = 0.764\end{aligned}$$

(d) Let's continue from Question 8.3(d) where we used a substitution of $u = 7x+5$. Find the new limits:

$$\begin{aligned}&\text{When } x = 0,\ u = 7\times 0+5 = 3 \\ &\text{When } x = 1,\ u = 7\times 1+5 = 12\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 \frac{1}{7x+5}dx &= \frac{1}{7}\int_3^{12}\frac{1}{u}du \\ &= -\frac{1}{7}\left[\ln(u)\right]_3^1 \\ &= \frac{1}{7}(\ln(12)-\ln(3)) = 0.198\end{aligned}$$

Question 8.5: Let's use $u = 2x+1$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 2x+1 \implies x = \frac{1}{2}(u-1) \implies \frac{dx}{du} = \frac{1}{2} \implies dx = \frac{1}{2}du$$

Find the new limits:

$$\begin{aligned}&\text{When } x = -1,\ u = 2\times(-1)+1 = -1 \\ &\text{When } x = 1,\ u = 2\times 1+1 = 3\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-1}^{1}\frac{4}{(2x+1)^3}dx &= \frac{1}{2}\int_{-1}^{3}\frac{1}{u^3}du \\ &= \frac{1}{2}\left[-\frac{1}{2}\times u^{-2}\right]_{-1}^{3} \\ &= -\frac{1}{4}(3^{-2}-(-1)^{-2}) = 0.222\end{aligned}$$

Question 8.6: We want to integrate the following:

$$\frac{dy}{dx} = (ax+b)^n$$

Let's use $u = ax+b$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = ax+b \implies x = \frac{1}{a}(u-b) \implies \frac{dx}{du} = \frac{1}{a} \implies dx = \frac{1}{a}du$$

Use these to make the substitution:

$$\int (ax+b)^n dx \implies \frac{1}{a}\int u^n du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{a}\int u^n du = \frac{1}{a(n+1)}u^{n+1}+c$$

and substituting $u = ax+b$ back in gives us:

$$y = \frac{(ax+b)^{n+1}}{a(n+1)}+c$$

You can differentiate using the chain rule to verify that this works.

Real-world problems

Question 8.7: We want to solve the hydrostatic equation to find the pressure P at $z = 1000$ when the surface pressure is $100,000$Pa (i.e. $z = 0$) and $T_{surface} = 290$. We rearrange and integrate pressure, P, between 100000 and P and z between 0 and 1000:

$$\begin{aligned}\frac{dP}{dz} &= -\frac{Pg}{R(T_{surface}-0.01z)} \\ \int_{100000}^{P}\frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{1000}\frac{1}{290-0.01z}dz\end{aligned}$$

Let's use $u = 290-0.01z$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 290-0.01z \implies x = 100(290-u) \implies \frac{dx}{du} = -100 \implies dx = -100du$$

Find the new limits

$$\begin{aligned}&\text{When } x=0,\ \ u=290-0.01\times 0=290\\&\text{When } x=1000,\ \ u=290-0.01\times(1000)=280\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{100000}^{P}\frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{1000}\frac{1}{290-0.01z}dz\\ \int_{100000}^{P}\frac{1}{P}dP &= \frac{100g}{R}\int_{290}^{280}\frac{1}{u}du\\ [\ln P]_{100000}^{P} &= \frac{100g}{R}[\ln u]_{290}^{280}\\ \ln(P)-\ln(100000) &= \frac{100g}{R}(\ln(280)-\ln(290))\\ \ln\left(\frac{P}{100000}\right) &= \frac{100g}{R}\ln\left(\frac{280}{290}\right)\end{aligned}$$

Rearranging this and substituting in $g=9.8$ and $R=287$, gives us

$$P=100000\times\exp\left(\frac{100\times 9.8}{287}\ln\frac{280}{290}\right)\cong 88708\text{Pa}$$

and thus the pressure at 1000m is 88708Pa.

Question 8.8: We can also demonstrate how to integrate the hydrostatic equation between general limits:

$$\begin{aligned}\frac{dP}{dz} &= -\frac{Pg}{R(T_{surface}-0.01z)}\\ \int_{P_0}^{P}\frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{z}\frac{1}{T_{surface}-0.01z}dz\end{aligned}$$

Let's use $u=T_{surface}-0.01z$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u=T_{surface}-0.01z\implies x=100(T_{surface}-u)\implies\frac{dx}{du}=-100\implies dx=-100du$$

Find the new limits

$$\begin{aligned}&\text{When } x=0,\ \ u=T_{surface}-0.01\times 0=T_{surface}\\&\text{When } x=1000,\ \ u=T_{surface}-0.01z\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}
\int_{P_0}^{P} \frac{1}{P} dP &= -\frac{g}{R} \int_0^z \frac{1}{290 - 0.01z} dz \\
\int_{P_0}^{P} \frac{1}{P} dP &= \frac{100g}{R} \int_{T_{surface}}^{T_{surface} - 0.01z} \frac{1}{u} du \\
[\ln P]_{P_0}^{P} &= \frac{100g}{R} [\ln u]_{T_{surface}}^{T_{surface} - 0.01z} \\
\ln(P) - \ln(P_0) &= \frac{100g}{R} (\ln(T_{surface} - 0.01z) - \ln(T_{surface})) \\
\ln\left(\frac{P}{P_0}\right) &= \frac{100g}{R} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)
\end{aligned}$$

Rearranging this and substituting in $g = 9.8$ and $R = 287$, gives us

$$P(z) = P_0 \exp\left(\frac{100 \times 9.8}{287} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

the desired expression.

Question 8.9: This Question 8.extends Question 8.6 from Chapter 7, however, the sphere has wind speed as well as the speed of the object. This makes the integration a little more difficult. The total forces on the object are

$$\pi r^2 \frac{\rho}{2} (v + w)^2$$

We want to estimate the distance the sphere with an initial velocity of $10m/s$ travels before slowing down to $0.01m/s$ with a wind speed of $w = 10$:

$$\begin{aligned}
mv\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2} (v + w)^2 \\
m\frac{v}{(v+w)^2}\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2} \\
m\frac{v}{(v+10)^2}\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2}
\end{aligned}$$

Using integration by substitution we can integrate this ODE. In terms of the limits, we want to estimate the distance travelled as the sphere goes from $10m/s$ and $0.01m/s$, so we integrate x between 0 and x, and integrate v from 10 to 0.01:

$$m \int_{10}^{0.01} \frac{v}{(v+10)^2} dv = -\int_0^x \pi r^2 \frac{\rho}{2} dx$$

Let's use $z = v + 10$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$z = v + 10 \implies v = z - 10 \implies \frac{dv}{dz} = 1 \implies dv = dz$$

Find the new limits:

$$\text{When } x = 10,\ \ z = 10 + 10 = 20$$
$$\text{When } x = 0.01,\ \ z = 0.01 + 10 = 10.01$$

Make the substitution with limits and integrate:

$$\begin{aligned}
m\int_{10}^{0.01} \frac{v}{(v+w)^2}dv &= -\int_0^x \pi r^2 \frac{\rho}{2}dx \\
m\int_{20}^{10.01} \frac{z-10}{z^2}dz &= -\int_0^x \pi r^2 \frac{\rho}{2}dx \\
m\int_{20}^{10.01} \frac{1}{z} - \frac{10}{z^2}dz &= -\left[\pi r^2 \frac{\rho}{2}\right]_0^x \\
m\left[\ln(z) + \frac{10}{z}\right]_{20}^{10.01} &= -\pi r^2 \frac{\rho}{2}x \\
m\left(\ln(10.01) + \frac{10}{10.01} - \ln(20) - \frac{10}{20}\right) &= -\pi r^2 \frac{\rho}{2}x \\
-0.193m &= -\pi r^2 \frac{\rho}{2}x
\end{aligned}$$

Substituting $m = 1$, $\rho = 1.2$, and $r = 0.5$ gives us:

$$\begin{aligned}
-0.193 &= -0.471x \\
x &= \frac{0.193}{0.471} = 0.41
\end{aligned}$$

So we see that the total distance travelled has reduced from 14.66 to 0.41m by adding in wind speed, so the wind causes the sphere to slow down much quicker.